

Wireless Communication Numerical Companion

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

Ch 6 · Speech and Channel Coding Numerical Problems & Solutions

The examined calculation here is the **7-bit Hamming code** (80 Ch 5), worked both ways below. **Convolutional encoding, Viterbi decoding** (HOT 5/22) are asked as “explain”, but they are only convincing with a worked trellis, so one is given. The remaining problems are the tutorial-set coder-rate calculations, which are the standard follow-ups to the speech coding theory.

The formula sheet

- **Hamming code** $n = 2^m - 1$, $k = 2^m - m - 1$, $n - k = m$, $d_{\min} = 3$, $t = 1$, $\text{rate} = k/n$
- **Detect t errors:** $d' \geq t + 1$. **Correct t errors:** $d' \geq 2t + 1$
- **Convolutional code** $\text{rate} = k/n$, constraint length K , 2^{K-1} trellis states
- **Nyquist** a band-pass signal of bandwidth B needs $f_s = 2B$ samples/s
- **Bit rate** $R = f_s \times (\text{bits per sample})$
- **Quantizer** $D = \int (x - f_Q(x))^2 p(x) dx$, $S = E[x^2] = \int x^2 p(x) dx$, $\text{SDR} = 10 \log_{10}(S/D)$
- **Coding gain** $G = \left(\frac{E_b}{N_0} \right)_{\text{uncoded}} - \left(\frac{E_b}{N_0} \right)_{\text{coded}}$ dB, at the same BER

Problem 1 7-bit Hamming code: encode and then detect

PIN 1/22 80 Ch 5 · the only pure channel-coding calculation in 22 papers, so it is worth banking exactly

QUESTION AS PRINTED 2080 Chaitra · Q7 · [3+5]

Explain the characteristics of Speech. Explain 7-bit hamming code method with relevant example.

The paper prints no data word, so supply your own. The run below encodes **1011** into a (7, 4) Hamming codeword, then receives a corrupted word and locates the faulty bit: that is the “relevant example” the 5 marks are for. **verified/added**

Step 0: the structure, once

- For $m = 3$: $n = 2^3 - 1 = 7$, $k = 2^3 - 3 - 1 = 4$, parity bits = 3.
- Parity bits go in positions that are **powers of 2**: 1, 2, 4. Data fills the rest: 3, 5, 6, 7.
- Layout, written with bit 7 on the *left* and bit 1 on the *right*:

$$D_7 \ D_6 \ D_5 \ P_4 \ D_3 \ P_2 \ P_1$$

- Each parity bit checks the positions whose binary index contains its own bit:

Parity bit	Checks positions	Binary indices	Why
P_1	1, 3, 5, 7	001, 011, 101, 111	all have bit 0 set
P_2	2, 3, 6, 7	010, 011, 110, 111	all have bit 1 set
P_4	4, 5, 6, 7	100, 101, 110, 111	all have bit 2 set

This is the whole trick. Because the groups are built on the binary indices, the three parity checks read together are the binary address of the faulty bit.

Part A Encode the data word 1011 with even parity

- **Step 1:** place the data bits into positions 7, 6, 5, 3:

$$D_7 = 1, \quad D_6 = 0, \quad D_5 = 1, \quad D_3 = 1$$

- **Step 2:** compute each parity bit so that its group has an **even** number of ones.

Parity	Group contents	Ones already present	Set to
P_1	$(P_1, D_3, D_5, D_7) = (P_1, 1, 1, 1)$	3 (odd)	1
P_2	$(P_2, D_3, D_6, D_7) = (P_2, 1, 0, 1)$	2 (even)	0
P_4	$(P_4, D_5, D_6, D_7) = (P_4, 1, 0, 1)$	2 (even)	0

- **Step 3: assemble the codeword, bit 7 first:**

Position	7	6	5	4	3	2	1
Contents	D_7	D_6	D_5	P_4	D_3	P_2	P_1
Value	1	0	1	0	1	0	1

transmitted codeword = 1010101

Part B A codeword 1011011 is received. Is it correct? If not, locate the error.

- **Step 1: lay out the received bits.**

Position	7	6	5	4	3	2	1
Received	1	0	1	1	0	1	1

- **Step 2: recheck each parity group, still assuming even parity.**

Check	Positions	Bits read	Parity	Syndrome bit
P_1	1, 3, 5, 7	1, 0, 1, 1	ODD → error	1
P_2	2, 3, 6, 7	1, 0, 0, 1	EVEN → ok	0
P_4	4, 5, 6, 7	1, 1, 0, 1	ODD → error	1

- **Step 3: read the syndrome.** Write it as $P_4P_2P_1$:

$$\text{syndrome} = 101_{(2)} = 4 + 0 + 1 = 5$$

- **Step 4: conclude.**

The received word is INCORRECT; bit position 5 is in error.

- **Step 5: correct it by flipping bit 5, 1 → 0:**

$$\text{corrected codeword} = 1001011$$

$$\text{recovered data (positions 7,6,5,3)} = 1000$$

- **Non-zero syndrome = error, and its decimal value is the position. A zero syndrome means no single-bit error.**
- **Limitation to state:** $d_{\min} = 3$, so this code corrects **one** bit and detects **two**. A **two-bit** error produces a non-zero syndrome pointing at the *wrong* position, and “correcting” it makes things worse. **That is why Hamming codes are always paired with interleaving** (Ch 5) on a fading channel, where errors arrive in bursts.

Hamming code parameters for other m

m	$n = 2^m - 1$	$k = 2^m - m - 1$	parity bits	rate k/n
3	7	4	3	0.571
4	15	11	4	0.733
5	31	26	5	0.839
6	63	57	6	0.905

The rate improves as m grows, but the code still corrects only **one** bit per block, so the protection per bit gets weaker. That trade-off is worth one line in any Hamming answer.

Problem 2 Convolutional encoding and Viterbi decoding

HOT 5/22 74 Bh **5**, 75 Bh, 70 Ma, 72 Ash **3**, 73 Bh **5**

QUESTION AS PRINTED 5 papers, always as a short note

Write short notes on: **Convolutional encoding and decoding** (2074 Bhadra Q9 c **5**)

Write short notes on: **Viterbi decoding algorithm** (2073 Bhadra Q9 c **5** · 2072 Ashwin Q9 c **3** · 2075 Bhadra Q12 b · 2070 Magh Q10 c)

PROBLEM not a PYQ · the example that carries the short note verified/added

For the rate $1/2$, constraint length $K = 3$ convolutional encoder defined by $X_1 = M + M_1 + M_2$ and $X_2 = M + M_2$ (mod-2 addition), encode the message **1011**. Then decode the received sequence $Y = 11\ 01\ 10$ using the Viterbi algorithm, showing the trellis and the surviving path metrics.

Part A Encoding 1011

- State = the contents of the two memory stages (M_1M_2), so there are $2^{K-1} = 4$ states: $a = 00, b = 01, c = 10, d = 11$.
- Two flush (tail) bits are appended so the encoder returns to state a .

Input M	State (M_1M_2)	$X_1 = M + M_1 + M_2$	$X_2 = M + M_2$	Output	Next state
1	00	$1+0+0 = 1$	$1+0 = 1$	11	10
0	10	$0+1+0 = 1$	$0+0 = 0$	10	01
1	01	$1+0+1 = 0$	$1+1 = 0$	00	10
1	10	$1+1+0 = 0$	$1+0 = 1$	01	11
0	11	$0+1+1 = 0$	$0+1 = 1$	01	01
0	01	$0+0+1 = 1$	$0+1 = 1$	11	00

encoded output = 11 10 00 01 01 11

- 4 message bits plus 2 flush bits give 6 input bits, and at rate $1/2$ that is **12 output bits**. ✓
- **Code rate including the tail** is $4/12 = 1/3$, not $1/2$. Real systems use long frames so the tail overhead is negligible. Worth a line.

Part B Viterbi decoding of $Y = 11\ 01\ 10$

- **Metric used: Hamming distance** between the branch's expected output and the received pair. Lower is better.
- At every node keep only the **smaller-metric** incoming path (the survivor) and discard the other.

Step	Received	State	Path metric	Surviving input path
1	11	00	2	0
		10	0	1
2	01	00	3	0, 0
		01	2	1, 0
		10	3	0, 1
		11	0	1, 1
3	10	00	3	1, 0, 0
		01	2	1, 1, 0
		10	3	1, 0, 1
		11	0	1, 1, 1

Reading the table

- **Step 1.** From state 00, input 0 gives output 00 (distance 2 from the received 11), input 1 gives output 11 (distance 0). So state 10 carries metric 0.
- **Step 2.** From state 10, input 0 gives 10 (distance 1 from 01) and input 1 gives 01 (distance 0). The path 1, 1 into state 11 keeps metric 0.
- **Step 3.** From state 11, input 1 gives output 10, which **matches the received 10 exactly**, so the metric stays 0.
- **Survivor:** the path ending in state 11 with total metric 0.

decoded message = 1 1 1

- **Verify by re-encoding** 1, 1, 1 from state 00: 11, then 01, then 10 = Y exactly. ✓ **Metric 0 means the received sequence was a valid codeword, ie no channel errors.**
- **Why Viterbi is tractable:** after every step there are still only 4 survivors, never more, because each state keeps one path. An exhaustive tree search would have 2^L paths after L bits.

Problem 3 Sub-band coder bit rate

PIN 1/22 80 Bb 6 · the calculation behind “sub-band coding with transmitter and receiver”

QUESTION AS PRINTED 2080 Bhadra · Q6 (a) · [6]

Explain sub-band coding with transmitter and receiver.

PROBLEM not a PYQ · the rate sum behind the block diagram verified/added

A sub-band coder splits the input into the four sub-bands tabulated below and samples each at its band-pass Nyquist rate. Find the total encoding bit rate, and compare it with coding the full band at the same resolution.

Sub-band	Band (Hz)	Bandwidth	$f_s = 2B$	Bits/sample	Rate (bps)
1	225–450	225 Hz	450	4	1800
2	450–900	450 Hz	900	3	2700
3	1000–1500	500 Hz	1000	2	2000
4	1800–2700	900 Hz	1800	1	1800
Total					8300

- **Step 1: bandwidth of each band** $= f_{\text{high}} - f_{\text{low}}$.
- **Step 2: sampling rate.** For **perfect reconstruction** each band-pass signal must be sampled at **twice its own bandwidth**, not twice its highest frequency. **That saving is the whole reason for the low-pass translation step in the sub-band coder.**

$$f_{s,1} = 2(450 - 225) = 450 \text{ samples/s}, \quad f_{s,2} = 2(900 - 450) = 900$$

$$f_{s,3} = 2(1500 - 1000) = 1000, \quad f_{s,4} = 2(2700 - 1800) = 1800$$

- **Step 3: total rate.**

$$R = 450(4) + 900(3) + 1000(2) + 1800(1)$$

$$= 1800 + 2700 + 2000 + 1800 = \boxed{8300 \text{ bps} = 8.3 \text{ kbps}}$$

- **Note the bit allocation:** the **narrowest, lowest** band gets the **most** bits (4), and the widest, highest band gets the fewest (1). That follows directly from the **PSD property** of speech: low bands carry most of the energy and most of the perceptual weight.
- **Compare:** plain 8 kHz / 8-bit PCM of the same speech would need **64 kbps**. Sub-band coding achieves nearly **8:1 compression** while remaining a waveform coder.

Problem 4 Upper bound on speech coder rate in an FDMA system

tutorial set; the calculation behind “choosing a speech codec for mobile communications”

PROBLEM not a PYQ · the sum behind “choosing a speech codec” verified/added

A digital mobile system uses 810–826 MHz for the forward channel and 940–956 MHz for the reverse channel. 90% of the bandwidth is given to traffic channels, and the system must support at least 1150 simultaneous calls using FDMA. The modulation achieves a spectral efficiency of 1.68 bps/Hz, and channel impairments force rate 1/2 forward error correction. Find the upper bound on the transmission bit rate that a speech coder in this system can use.

- **Step 1: total band.**

$$B_{\text{total}} = 826 - 810 = 16 \text{ MHz}$$

The reverse band is the same width, and FDMA pairs them, so work with one direction.

- **Step 2: bandwidth actually available to traffic.**

$$B_{\text{traffic}} = 0.9 \times 16 = 14.4 \text{ MHz}$$

The other 10 % goes to control channels and guard bands.

- **Step 3: bandwidth per channel.** FDMA divides it among the simultaneous calls:

$$B_{\text{ch}} = \frac{14.4 \text{ MHz}}{1150} = 12.52 \text{ kHz}$$

$$\Rightarrow \boxed{\text{maximum channel bandwidth} = 12.5 \text{ kHz}}$$

- **Step 4: raw bit rate the channel can carry.**

$$R_{\text{raw}} = 1.68 \frac{\text{bps}}{\text{Hz}} \times 12.52 \text{ kHz} = 21.04 \text{ kbps}$$

- **Step 5: subtract the FEC overhead.** A rate 1/2 code doubles the bits, so only half the raw rate is available to the speech coder:

$$R_{\text{speech}} = 0.5 \times 21.04 = \boxed{10.52 \text{ kbps}}$$

- **Interpretation:** the speech coder must run at **10.5 kbps or below**. A plain waveform coder cannot do that at toll quality, so this system needs a **hybrid or vocoder** class coder, eg RPE-LTP at 13 kbps (too fast here) or a CELP variant at 8 kbps (fits).
- **This is exactly the reasoning that forces cellular systems onto low-rate coders**, and it is the numerical answer to “why do we need speech coding techniques”.

Problem 5 Quantizer mean square error and signal to distortion ratio

tutorial set; the calculation behind the PDF property of speech **TOP 7/22**

PROBLEM not a PYQ · the PDF property of speech, made numerical **verified/added**

The input signal to a quantizer has the probability density function $p(x) = x/32$ for $0 \leq x \leq 8$ and zero elsewhere. The quantization levels are $\{1, 3, 5, 7\}$.

- Compute the mean square error distortion at the quantizer output.
- Compute the output signal to distortion ratio.
- How would you redistribute the quantization levels to decrease the distortion, and for what input pdf would this quantizer already be optimal?

- **Step 0: check the pdf is valid.**

$$\int_0^8 \frac{x}{32} dx = \left[\frac{x^2}{64} \right]_0^8 = \frac{64}{64} = 1 \quad \checkmark$$

- **Step 1: set the decision boundaries.** With levels at 1, 3, 5, 7 the midpoints give boundaries at $\{0, 2, 4, 6, 8\}$, so the quantizer is **uniform** with step 2.
- **Step 2: mean square error distortion.**

$$\begin{aligned} D &= \int_{-\infty}^{\infty} (x - f_Q(x))^2 p(x) dx \\ &= \int_0^2 (x - 1)^2 \frac{x}{32} dx + \int_2^4 (x - 3)^2 \frac{x}{32} dx + \int_4^6 (x - 5)^2 \frac{x}{32} dx + \int_6^8 (x - 7)^2 \frac{x}{32} dx \end{aligned}$$

- **Step 3: evaluate one integral to show the method.** For the first,

$$\begin{aligned} \int_0^2 (x^2 - 2x + 1) \frac{x}{32} dx &= \frac{1}{32} \int_0^2 (x^3 - 2x^2 + x) dx \\ &= \frac{1}{32} \left[\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} \right]_0^2 = \frac{1}{32} \left(4 - \frac{16}{3} + 2 \right) = \frac{1}{32} \cdot \frac{2}{3} = \frac{1}{48} \end{aligned}$$

The other three evaluate to $\frac{3}{48}$, $\frac{5}{48}$ and $\frac{7}{48}$ by the same method.

- **Step 4: total distortion.**

$$D = \frac{1 + 3 + 5 + 7}{48} = \frac{16}{48} = \frac{1}{3} = \boxed{0.3333}$$

- **Step 5: signal power.**

$$S = E[x^2] = \int_0^8 x^2 \frac{x}{32} dx = \frac{1}{32} \left[\frac{x^4}{4} \right]_0^8 = \frac{1}{32} \cdot \frac{4096}{4} = \boxed{32}$$

- **Step 6: signal to distortion ratio.**

$$\text{SDR} = 10 \log_{10} \left(\frac{S}{D} \right) = 10 \log_{10} \left(\frac{32}{0.3333} \right) = 10 \log_{10}(96.0) = \boxed{19.82 \text{ dB}}$$

- **Step 7: how to reduce the distortion.**

- Note that the four sub-integrals came out as 1, 3, 5, 7 over 48: **the top interval contributes seven times as much distortion as the bottom one.**
- That is because $p(x) = x/32$ rises with x , so most of the probability mass sits at large amplitudes.
- **Fix: use a non-uniform quantizer.** Place **more levels where the probability is high** (large x) and fewer where it is low (small x), ie shrink the step size towards $x = 8$.
- This is exactly the **PDF property** of speech being exploited, and it is the principle behind μ -law and A-law companding.
- **Step 8: for what pdf is this quantizer optimal?** A uniform quantizer is optimal when every interval contributes equally, which happens only when the input is **uniformly distributed** over $[0, 8]$, ie $p(x) = 1/8$.

Problem 6 Speech coder rates: what fits in what

supporting numbers for the vocoder **TOP 10/22** and LPC **TOP 8/22** questions

Reference figures. **Quoting one correct rate is often worth a mark** in an otherwise descriptive answer.

Coder	Class	Bit rate	Note
PCM (toll reference)	Waveform	64 kbps	8 kHz, 8 bit, μ -law
ADPCM	Waveform	32 kbps	G.726
Sub-band coding	Waveform	9.6–32 kbps	4 or 8 bands
Adaptive transform (ATC)	Waveform	9.6–20 kbps	DCT based
Voice-excited vocoder	Hybrid	7.2–9.6 kbps	no pitch extraction needed
GSM RPE-LTP	Hybrid	13 kbps	full rate GSM
CELP	Vocoder	4.8 kbps	codebook excitation
LPC	Vocoder	4.8 kbps (good quality), 2.4 kbps	all-pole model

- **Compression achieved** against the 64 kbps PCM reference: ADPCM $2\times$, sub-band $\approx 7\times$, GSM $\approx 5\times$, CELP $\approx 13\times$.
- **The pattern to remember:** rate falls, complexity rises, and robustness falls. A 4.8 kbps CELP frame damaged by the channel is far more audible than a damaged 64 kbps PCM sample. **That is why the lowest-rate coders need the strongest channel coding**, which is the link between the two halves of this chapter.

Problem 7 Coding gain

syllabus, no PYQ yet · included because it is the number that justifies every code in the chapter

PROBLEM not a PYQ · the number that justifies every code here **verified/added**

An uncoded BPSK link needs $E_b/N_0 = 10.5$ dB to reach a BER of 10^{-5} . With a rate $1/2$, $K = 7$ convolutional code and soft-decision Viterbi decoding, the same BER is reached at $E_b/N_0 = 4.5$ dB. Find the coding gain in dB and the transmit power saving it represents.

- **Step 1: coding gain.**

$$G = \left(\frac{E_b}{N_0} \right)_{\text{uncoded}} - \left(\frac{E_b}{N_0} \right)_{\text{coded}} = 10.5 - 4.5 = \boxed{6 \text{ dB}}$$

- **Step 2: what 6 dB is worth in power.**

$$10^{6/10} = 3.98 \approx 4$$

The transmitter can run at one quarter of the power for the same quality.

- **Step 3: what it is worth in range**, at path loss exponent $n = 3.5$:

$$10^{6/(10 \times 3.5)} = 10^{0.1714} = 1.48$$

ie **48 % more range**, or $1.48^2 = 2.2$ times the area from one site.

- **Step 4: what it costs.** The rate $1/2$ code **doubles the transmitted bit rate**, so it needs twice the bandwidth (or halves the user data rate).
- **TCM exists precisely to escape that cost:** by combining coding with a larger signal constellation it delivers coding gains up to 6 dB with **no bandwidth expansion**. Turbo codes then push the gain further still, close to the Shannon bound.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
WLAN	wireless local area network	WMAN	wireless metropolitan area network
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM

evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.